Python Basics

ASSIGNMENT

Question 1 : What is Python and why is it widely used in Data Analytics? Mention any three reasons.

ANS- Python is a high-level, interpreted programming language that makes it easy for anyone — to learn and use. It's known for its simple syntax, readability, and versatility — meaning it can be used for almost anything: from building websites to analyzing data or training AI models **.**

It is widely used in Data Analytics;-

- Beginner Friendly as Python has simple, readable syntax.
- Also it is a good start for non programmers. Easy Syntax as Python uses natural language structure where indentation makes the code readable.
- Python has Large Community Support.
- Powerful Libraries like Pandas, NumPy, Seaborn, Matplotlib, Scikit-learn.

Question 2 : Explain the difference between List and Tuple in Python

ANS- List is mutable (can be changed) and uses square brackets [ ], while Tuple is immutable (cannot be changed) and uses parentheses ( ).

```
list1 = [1, 2, 3]     # can be modified
tuple1 = (1, 2, 3)    # cannot be modified
```

Question 3 : What is a function in Python? Why are functions useful?

ANS- A function in Python is a block of reusable code that performs a specific task. It is defined using the keyword.

Functions are useful because:

- They avoid repetition by reusing code.
- They make programs easier to read and maintain.
- They allow breaking complex problems into smaller parts.

Question 4 : Write a Python program to take a user's name as input and print a greeting message.

ANS-A Python program to take a user's name as input and print a greeting message can be written like this:

```
#program to greet user
```

```
name = input("Enter your name: ")

print("Hello, " + name + "! Welcome to Python.")
```

Question 5 : Write a Python program to check whether a number is even or odd using conditional statements.

ANS- # Program to check even or odd

```
num = int(input("Enter a number: "))

if num % 2 == 0:

    print("The number is Even")

else:

    print("The number is Odd")
```

Output IF number is 8

The number is Even

Output IF number is 7

The number is Odd

Question 6 : Write a program to print numbers from 1 to 10 using a loop

ANS- # Program to print numbers from 1 to 10

```
for i in range(1, 11):

    print(i)
```

- `range(1, 11)` generates numbers starting at **1** and ending just before **11** (so it stops at 10).
- The `for` loop goes through each number in that range.
- `print(i)` displays the current number on the screen.

OUTPUT

1 2 3 4 5 6 7 8 9 10

Question 7 : Create a list of five numbers and print the maximum number from the list.

ANS- # Program to find maximum number in a list

numbers = [12, 45, 7, 23, 89]

print("The maximum number is:", max(numbers))

- `numbers = [12, 45, 7, 23, 89]` creates a list with five elements.
- The `max()` function scans through the list and returns the largest value.
- `print()` displays the maximum number to the user.

OUTPUT

The maximum number is: 89

Question 8 : Write a Python program to remove duplicate values from a list using a set.

ANS- # Program to remove duplicates using a set

numbers = [10, 20, 30, 20, 40, 10]

unique_numbers = list(set(numbers))

print("List after removing duplicates:", unique_numbers)

- `numbers` is the original list with duplicates (`10` and `20` appear twice).
- `set(numbers)` converts the list into a set, which keeps only unique values.
- `list(set(numbers))` changes the set back into a list format.

Finally, `print()` displays the new list without duplicates.

OUTPUT

List after removing duplicates: [40, 10, 20, 30]

Question 9 : Write a function that returns the square of a number.

ANS- # Function to return square of a number

```
def square(num):
    return num * num


# Example
result = square(5)
print("Square is:", result)
```

- `def square(num):` defines a function named `square` that takes one parameter `num`.
- `return num * num` sends back the result of multiplying the number by itself.
- When you call `square(5)`, the function returns `25`.
- The `print()` statement displays the result.

OUTPUT

```
Square is: 25
```

Question 10 : Write a Python program to count how many times a number appears in a list. Example List [2,3,4,2,5,2]

ANS-# Program to count occurrences of a number in a list

```
numbers = [2, 3, 4, 2, 5, 2]
count_of_2 = numbers.count(2)
print("Number 2 appears", count_of_2, "times")
```

- `numbers` is the list given in the example.
- `numbers.count(2)` checks how many times the value `2` occurs in the list.
- The result is stored in `count_of_2`.
- `print()` displays the count.

OUTPUT

```
Number 2 appears 3 times
```